



INTERNATIONAL MATHEMATICS SUMMER CAMP IMSC25

POLYNOMIALS

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1. Given a polynomial P with integer coefficients, we define the polynomial $f(P)$ as follows:

-If P is a monic polynomial with degree $n \geq 1$ and n integer roots $r_0 \leq r_1 \leq \dots \leq r_{n-1}$ (counted with multiplicity) then

$$(f(P))(x) = x^n + r_{n-1}x^{n-1} + r_{n-2}x^{n-2} + \dots + r_1x + r_0$$

-Otherwise, $f(P)$ is the zero polynomial.

Find all nonconstant monic polynomials P with integer coefficients such that the sequence $P, f(P), f(f(P)), \dots$ does not contain the zero polynomial.

2. Let n be a positive integer, and let f, g and h denote polynomials with real coefficients whose degrees are at most n . What is the largest possible number of real numbers x for which $f(x), g(x), h(x)$ in some order, are consecutive terms of a non-constant arithmetic sequence, assuming that there are only a finite number of such x ?

3. Real polynomial of odd degree has all positive coefficients. Prove that there is a (possibly trivial) permutation of the coefficients such that the resulting polynomial has exactly one real zero.

4. Let $P \in \mathbb{R}[x]$. Suppose that the multiset of real roots (where roots are counted with multiplicity) of $P(x) - x$ and $P^3(x) - x$ are distinct. Prove that for all $n \in \mathbb{N}$, $P^n(x) - x$ has at least $\sigma(n) - 2$ distinct real roots.

(Here $P^n(x) := P(P^{n-1}(x))$ with $P^1(x) = P(x)$, and $\sigma(n)$ is the sum of all positive divisors of n).

5. Let us choose arbitrarily n vertices of a regular $2n$ -gon and color them red. The remaining vertices are colored blue. We arrange all red-red distances into a non-decreasing sequence and do the same with the blue-blue distances. Prove that the sequences are equal.

6. Let $\mathbb{R}[x]$ be the set of all polynomials with real coefficients. Find all functions $f : \mathbb{R}[x] \rightarrow \mathbb{R}[x]$ satisfying the following conditions:

- f maps the zero polynomial to itself,
- for any non-zero polynomial $P \in \mathbb{R}[x]$, $\deg f(P) \leq 1 + \deg P$, and

• for any two polynomials $P, Q \in \mathbb{R}[x]$, the polynomials $P - f(Q)$ and $Q - f(P)$ have the same set of real roots.

7. Prove that for any polynomial P with real coefficients, and for any positive integer n , there exists a polynomial Q with real coefficients such that $P(x)^2 + Q(x)^2$ is divisible by $(1 + x^2)^n$.

8. N different numbers are written on blackboard and one of these numbers is equal to 0. One may take any polynomial such that each of its coefficients is equal to one of written numbers (there may be some equal coefficients) and write all its roots on blackboard. After some of these operations all integers between -2016 and 2016 were written on blackboard (and some other numbers maybe). Find the smallest possible value of N .

9. Given $n \geq 3$ distinct points $P_1, P_2, \dots, P_n$ on the unit circle. If it's known that for any other point Q on the unit circle, one has $\prod_{i=1}^n |QP_i| \leq 2$. Show that those n points are vertices of a regular n -gon.

10. Given is a monic polynomial f of degree n with real coefficients and integers $x_0 < x_1 < \dots < x_n$. Prove that there exist some positive integer k , such that $|f(x_k)| \geq \frac{n!}{2^n}$.

11. We call two non-constant polynomials friendly if each of them has only real roots, and every root of one polynomial is also a root of the other. For two friendly polynomials $P(x), Q(x)$ and a constant $C \in \mathbb{R}, C \neq 0$, it is given that $P(x) + C$ and $Q(x) + C$ are also friendly polynomials. Prove that $P(x) \equiv Q(x)$.

12. Assume that the coefficients of the polynomial $P(x) = \sum_{k=0}^d a_k x^k$ are positive, and for all $k = 1, 2, \dots, d-1$, $a_k^2 > 9a_{k+1}a_{k-1}$. Prove that $P(x)$ have d different real roots.

13. Determine all positive integers n for which there exist pairwise distinct integers $a_1, \dots, a_n$ and $b_1, \dots, b_n$ such that

$$\prod_{i=1}^n (a_k^2 + a_i a_k + b_i) = \prod_{i=1}^n (b_k^2 + a_i b_k + b_i) = 0, \quad \forall k = 1, \dots, n.$$

14. The polynomials $P(x)$ and $Q(x)$ are given. It is known that for a certain polynomial $R(x, y)$ the identity $P(x) - P(y) = R(x, y)(Q(x) - Q(y))$ applies. Prove that there is a polynomial $S(x)$ so that $P(x) = S(Q(x)) \quad \forall x$.

15. Given a real number $t \geq 3$, suppose a polynomial $f \in \mathbb{R}[x]$ satisfies

$$|f(k) - t^k| < 1, \quad \forall k = 0, 1, \dots, n.$$

Prove that $\deg f \geq n$.

16. Let $P, Q \in \mathbb{R}[x]$ be relatively prime nonconstant polynomials. Show that there can be at most three real numbers λ such that $P + \lambda Q$ is the square of a polynomial.